INTERNSHIP REPORT

On

Leaf Detection and Soybean Disease Classification Using a Two-Stage Deep Learning Pipeline

BY

SVN Sai Sathvik, Kotyada Parthiv

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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AN INTERNSHIP REPORT

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Submitted by:

SVN Sai Sathvik, Kotyada Parthiv

Internship Supervisor:

Sourab

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Preface

This internship report on “Leaf Detection and Soybean Disease Classification Using a Two-Stage Deep Learning Pipeline” documents the work carried out over a two-month internship period under the guidance of **Sourab**.

Through this report we have tried to give a detailed account of a two-stage deep learning system designed to improve the reliability and precision of automated soybean disease detection. The first stage employs a ResNet-based binary classifier to filter irrelevant non-leaf images, while the second stage uses a MobileNetV2-based model to classify confirmed leaf images into four agronomically relevant health categories.

We have tried, to the best of our abilities and knowledge, to explain the system design, datasets, experimental methodology, and results in a clear and methodical manner. Figures, tables, and quantitative metrics have been included to make the exposition as illustrative as possible.

SVN Sai Sathvik, Kotyada Parthiv
Interns, Department of Computer Science & Engineering
International Institute of Information Technology Bangalore

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We are also grateful to the broader open-source community and the contributors of the Kaggle datasets used in this study — particularly the *Leaf vs. Non-Leaf Images Dataset* and the *Soybean Leaf Image Dataset* — without which the experimental work would not have been possible.

Without the support of our supervisor and the resources made available by IIIT Bangalore, this internship project and report would not have been feasible.

SVN Sai Sathvik, Kotyada Parthiv
Department of Computer Science & Engineering
International Institute of Information Technology Bangalore

Abstract

Automated plant disease detection is a critical enabler of precision agriculture, reducing reliance on manual inspection and enabling timely intervention. Existing single-stage classifiers suffer from spurious activations when presented with non-leaf inputs — soil backgrounds, stems, hands, or unrelated objects — that should never reach the disease model. This internship project addresses that gap by proposing a **two-stage deep learning pipeline** specifically targeting soybean leaf disease classification.

Stage 1 employs a ResNet-based binary image classifier to filter inputs, accepting only confirmed leaf images and rejecting all irrelevant content. Trained and evaluated on the Kaggle *Leaf vs. Non-Leaf Images* dataset with augmentation, the Stage-1 model achieves a validation accuracy of **99.07%**, providing a high-confidence pre-filter.

Stage 2 applies a MobileNetV2-based multi-class classifier to the filtered leaf images, distinguishing among four target categories: *Healthy*, *Yellow Mosaic*, *Rust*, and *Sudden Death Syndrome (SDS)*. Trained on the Kaggle *Soybean Leaf Image Dataset* with the Adam optimiser (10 epochs, batch size 32), the Stage-2 model reaches a test accuracy of **86.32%**, a weighted F1 score of **0.856**, and a validation accuracy of **87.10%**.

The combined pipeline reduces wasted inference, improves system precision, and is lightweight enough for edge deployment on mobile and embedded agricultural devices. Future work will focus on expanding field-captured data, confusion-matrix-guided class-level refinement, and prototype deployment on embedded hardware.

Keywords: Soybean disease classification, leaf detection, ResNet, MobileNetV2, two-stage pipeline, precision agriculture, deep learning.

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CHAPTER 1

Introduction

1.1 Background and Motivation

Agriculture feeds the world, yet crop losses caused by plant diseases remain one of the most persistent threats to global food security. Soybeans (*Glycine max*) are among the most economically significant legume crops worldwide, supplying protein, oil, and animal feed to billions of people. Despite their importance, soybean plants are vulnerable to a range of fungal, viral, and bacterial diseases — including *Yellow Mosaic Virus*, *Rust* (*Phakopsora pachyrhizi*), and *Sudden Death Syndrome* — each of which can cause yield losses in excess of 30% if left undiagnosed [1].

Traditional disease diagnosis relies on visual inspection by trained agronomists, an approach that is time-consuming, expensive, and not scalable to the field conditions faced by smallholder farmers. Advances in deep learning and computer vision have created opportunities to automate this process through image-based classification systems deployable on low-cost, field-portable hardware.

1.2 Problem Statement

A naive approach to automated disease detection is to train a single multi-class classifier and let it distinguish among disease categories. This design has a critical weakness: it is fragile to *out-of-distribution inputs* such as photographs of soil, hands, stems, or ambient objects that inevitably appear in field-captured images. A classifier trained only on leaf images will assign one of its disease labels to any input, regardless of whether the input contains a leaf, leading to high false-positive rates in production.

This internship project addresses this weakness by decomposing the task into two sequential stages:

1. **Stage 1 — Leaf Detection:** A binary classifier determines whether an image contains a leaf. Non-leaf images are rejected before reaching Stage 2.

2. **Stage 2 — Disease Classification:** A multi-class classifier operates only on confirmed leaf images and assigns one of four health labels.

1.3 Objectives

The principal objectives of this internship project are:

- To design and train a high-accuracy binary leaf detector capable of reliably separating leaf from non-leaf images.

- To design and train a lightweight multi-class disease classifier for soybean leaves covering four agronomic categories.

- To integrate both stages into a coherent inference pipeline suitable for edge deployment.

- To evaluate both stages quantitatively using standard classification metrics (accuracy, precision, recall, F1 score) and identify avenues for future improvement.

1.4 Scope and Limitations

The current work is limited to four disease/health categories available in the chosen Kaggle dataset and does not yet generalise to the full spectrum of soybean diseases. All experiments were conducted on pre-curated, laboratory-augmented data; robustness under uncontrolled field conditions remains an open question addressed in the future-work section.

1.5 Report Organisation

The remainder of this report is organised as follows. Chapter 2 surveys related work in plant disease detection. Chapter 3 describes the datasets and preprocessing pipelines. Chapter 4 presents the system architecture and model design decisions. Chapter 5 details training configurations and reports experimental results. Chapter 6 summarises conclusions and outlines future work.

CHAPTER 2

Literature Review

2.1 Deep Learning for Plant Disease Detection

The use of deep convolutional neural networks (CNNs) for plant disease identification was brought into focus by the landmark study of Mohanty *et al.* [2], who trained a deep CNN on the PlantVillage dataset — containing over 54,000 images across 26 diseases and 14 crop species — and achieved a top-1 accuracy of approximately 99.35% under controlled laboratory conditions. Their work demonstrated that off-the-shelf CNN architectures, when trained on sufficiently large and diverse datasets, can match the diagnostic accuracy of trained agronomists.

Subsequent research explored the gap between laboratory performance and field accuracy. Ramcharan *et al.* [3] studied cassava disease detection using transfer learning from the Inception network and noted significant performance degradation when transitioning from curated to field-collected images, motivating augmentation-heavy pipelines and robustness engineering strategies.

2.2 Transfer Learning and Lightweight Architectures

Transfer learning — the practice of initialising a network with weights pre-trained on a large corpus (typically ImageNet) and fine-tuning on the target task — has become the default paradigm for agricultural image classification due to the limited size of most crop-disease datasets. Architectures such as VGG [4], ResNet [5], and MobileNet [6] have all been successfully applied.

For edge deployment, MobileNetV2 [7] — which employs depthwise separable convolutions and inverted residuals — offers an attractive tradeoff between accuracy and model size, achieving competitive performance on ImageNet at a fraction of the computational cost of full-size networks.

2.3 Two-Stage and Hierarchical Classification

Hierarchical and multi-stage classification systems have been explored in domains where out-of-distribution rejection is critical. Geetharamani and Pandian [9] employed a nine-layer CNN for a 38-class plant disease problem and showed that intermediate gating mechanisms can improve precision. The present work adopts a two-stage sequential gate — an idea related to the "reject option" in classifier design — specifically applied to the leaf/non-leaf discrimination problem.

2.4 YOLO-Based Approaches

Object-detection frameworks such as YOLO (You Only Look Once) [8] have been used for leaf localisation when the precise bounding box of the diseased region is required by downstream processes. These models are computationally heavier than classification-only approaches. Our Stage-1 comparison (Section 4.2.2) explicitly evaluates YOLO against ResNet for the leaf detection sub-task and shows that a classification model suffices and outperforms YOLO when only presence/absence detection is required.

2.5 Summary

The reviewed literature establishes three key design principles adopted in this work: (i) transfer learning from ImageNet pre-trained weights, (ii) lightweight architectures (MobileNetV2) for deployment feasibility, and (iii) two-stage pipeline design to handle out-of-distribution inputs gracefully.

CHAPTER 3

Datasets and Preprocessing

3.1 Stage 1 Dataset: Leaf vs. Non-Leaf Images

3.1.1 Source and Structure

The Stage-1 binary classifier was trained on the *Leaf vs. Non-Leaf Images Dataset* available on Kaggle [10]. The dataset contains two labelled classes:

- **Leaf:** Images containing one or more plant leaves, captured under varied lighting conditions, backgrounds, and angles.

- **Non-Leaf:** Images of diverse non-plant objects including soil, roads, hands, household items, and random textures.

The diversity of backgrounds and lighting conditions encourages the model to generalise to the noisy field conditions expected in production deployment.

3.1.2 Preprocessing Pipeline

All images were subjected to the following preprocessing steps:

1. **Resize:** Images were resized to 224×224 pixels.

2. **Normalisation:** Pixel values were normalised using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

3. **Augmentation:** Random horizontal flip and random rotation were applied during training to improve generalisation.

4. **Balanced sampling:** Class counts were balanced to avoid bias towards indoor-dominated non-leaf images.

3.2 Stage 2 Dataset: Soybean Leaf Image Dataset

3.2.1 Source and Structure

The Stage-2 disease classifier was trained on the *Soybean Leaf Image Dataset* available on Kaggle [11]. Only the following four target classes were retained from the full dataset:

Table 3.1: Target disease classes used from the Soybean Leaf Image Dataset

Class ID	Class Name	Description
0	Healthy	Normal leaf tissue; no visible disease symptoms
1	Yellow Mosaic	Viral-like mosaic patches and chlorosis across the leaf blade
2	Rust	Fungal pustules and bronzing on leaf underside and margins
3	Sudden Death Syndrome	Necrotic streaking and defoliation symptoms

3.2.2 Preprocessing Pipeline

1. **Class filtering:** Only the four target classes were retained.
2. **Label hygiene:** Inconsistent labels were resolved and corrupted image files were removed prior to training.
3. **Resize:** All images were resized to 224×224 pixels.
4. **Augmentation:** Random horizontal flip and random rotation were applied to increase robustness.
5. **Normalisation:** ImageNet-standard normalisation was applied.

3.2.3 Dataset Split

Both datasets were divided into training, validation, and test subsets using a stratified split to preserve class proportions across all three partitions.

CHAPTER 4

Methodology

4.1 System Architecture Overview

The complete inference pipeline is illustrated conceptually in Figure 4.1. An input image first passes through the Stage-1 leaf detector. If the detector classifies the image as “non-leaf”, the pipeline halts and returns a rejection response. Only images classified as “leaf” proceed to the Stage-2 disease classifier, which returns one of the four disease/health labels.

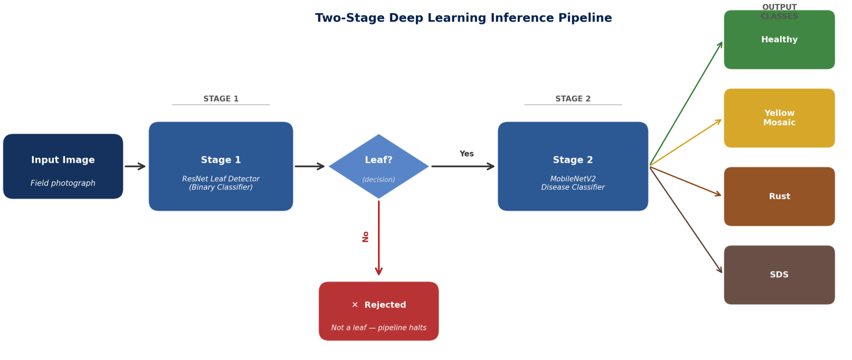

Figure 4.1: Two-stage deep learning pipeline for leaf detection and soybean disease classification.

This architecture provides three practical advantages:

- **Reduced wasted inference:** Compute is applied to Stage 2 only when an image is confirmed to contain a leaf.
- **Improved system precision:** Spurious disease predictions on non-leaf inputs are eliminated.

- **Edge-deployment feasibility:** Both stage models are lightweight enough for embedded or mobile deployment.

4.2 Stage 1 — Leaf Detection Model

4.2.1 Architecture: ResNet

ResNet (Residual Network) [5] was selected for Stage 1. ResNets introduce shortcut connections that allow gradients to flow directly through the network, alleviating the vanishing-gradient problem and enabling stable training of deeper networks. Key benefits are:

- Strong feature extraction from hierarchical residual blocks.
- Stability during fine-tuning via transfer learning.
- Efficient inference relative to the classification accuracy achieved.

The model was initialised with ImageNet pre-trained weights. The final fully connected layer was replaced with a two-class output head (leaf / non-leaf), and the network was fine-tuned end-to-end.

4.2.2 Model Selection: ResNet vs. YOLO

An alternative candidate for Stage 1 was YOLO, an object-detection framework capable of both localising and classifying leaves via bounding boxes. Table 4.1 summarises the comparison.

Table 4.1: ResNet vs. YOLO for leaf detection (Stage 1)

Criterion	**YOLO**	**ResNet (Selected)**
Task type	Detection + localisation	Binary classification
Output	Bounding boxes + labels	Single class probability
Compute cost	Higher	Lower
Validated accuracy	$\approx 94\%$	$\approx 99.07\%$
Edge suitability	Limited	High

ResNet outperforms YOLO on both accuracy and compute efficiency for this sub-task and was therefore selected for Stage 1.

4.3 Stage 2 — Disease Classification Model

4.3.1 Architecture: MobileNetV2

MobileNetV2 [7] was selected for Stage 2. It employs *depthwise separable convolutions* combined with *inverted residual blocks* to achieve efficient feature extraction at a fraction of the parameter count of standard convolutional networks.

Efficiency: Low parameter count and fast inference latency, enabling real-time use on mobile processors.

Strong Feature Representation: Depthwise separable convolutions preserve discriminative power for fine-grained texture and colour differences between disease classes.

Edge Readiness: Directly compatible with TFLite and ONNX exportation for embedded deployment.

The final classifier head was replaced with a four-class linear layer (Healthy, Yellow Mosaic, Rust, SDS), and the backbone was fine-tuned.

4.3.2 Loss Function and Optimiser

- **Loss:** Cross-Entropy Loss.
- **Optimiser:** Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$).
- **Batch size:** 32.
- **Epochs:** 10 (conservative budget; see Section 5.2.4).

CHAPTER 5

Experiments and Results

5.1 Stage 1 Results: Leaf Detector

5.1.1 Quantitative Performance

The ResNet-based leaf detector achieved a **validation accuracy of 99.07%**, demonstrating highly reliable binary discrimination between leaf and non-leaf inputs.

Table 5.1: Stage 1 — Leaf Detection Performance Summary

Metric	**Value**
Architecture	ResNet (ImageNet pre-trained)
Task	Binary classification (leaf / non-leaf)
Validation Accuracy	99.07%

5.1.2 Qualitative Testing

Visual tests were conducted by passing randomly selected green objects (potted plants, green plastic items, vegetables) through the model. The model correctly identified all leaf images as “leaf” and all non-leaf green objects as “non-leaf”, demonstrating that the classifier has learned leaf-specific structural features rather than merely detecting the colour green.

Figure 5.1 shows five representative test predictions made by the Stage-1 leaf detector. Each sub-figure caption reports the true label and the model’s predicted label.

Figure 5.1: Stage-1 leaf detector: five representative test predictions. Each image contains the model's prediction and ground truth label.

5.2 Stage 2 Results: Disease Classifier

5.2.1 Training Configuration

Table 5.2: Stage 2 — MobileNetV2 Training Configuration

Hyperparameter	**Value**
Input resolution	224×224 pixels
Augmentation	Random horizontal flip, random rotation
Normalisation	ImageNet statistics
Optimiser	Adam
Loss function	CrossEntropyLoss
Batch size	32
Number of epochs	10
Pre-training	ImageNet weights

5.2.2 Test Set Performance

The Stage-2 model was evaluated on a held-out test set of 95 images. Overall metrics are shown in Table 5.3 and per-class results in Table 5.4.

Table 5.3: Stage 2 — Overall Test Metrics (MobileNetV2, $n = 95$)

Metric	**Value**
Test Accuracy	86.32%
Macro Precision	0.8815
Macro Recall	0.8632
Macro F1 Score	0.8565
Validation Loss	0.1173
Validation Accuracy	87.10%

Table 5.4: Stage 2 — Per-Class Classification Report

Class	Precision	Recall	F1	Support
Healthy (0)	0.97	1.00	0.98	29
Yellow Mosaic (1)	0.92	0.55	0.69	22
Rust (2)	0.70	0.95	0.81	22
SDS (3)	0.91	0.91	0.91	22
Macro avg	0.87	0.85	0.85	95
Weighted avg	0.88	0.86	0.86	95

5.2.3 Discussion of Per-Class Results

Healthy (Class 0): The model performs near-perfectly (F1 = 0.98, recall = 1.00), indicating that healthy leaf images are visually distinct from all diseased categories.

Yellow Mosaic (Class 1): This class shows the weakest recall (0.55), indicating that roughly 45% of Yellow Mosaic instances are misclassified — most likely confused with the Rust class due to overlapping chlorotic patterns. Targeted augmentation and additional data collection are the primary improvement opportunities.

Rust (Class 2): Precision is moderate (0.70) but recall is high (0.95), suggesting a slight tendency to over-predict Rust when confronted with ambiguous cases from adjacent classes.

Sudden Death Syndrome (Class 3): Balanced precision and recall (both 0.91) indicate the model has learned reliable discriminative features for this class.

5.2.4 Overfitting Avoidance Strategy

Training was deliberately halted at 10 epochs. The validation loss of 0.1173 and validation accuracy of 87.10% closely mirror the test results (86.32%), confirming the model has not overfit. This conservative budget was chosen to encourage learning of broad disease patterns rather than memorising dataset-specific artefacts.

5.3 Qualitative Predictions: Disease Classifier

Figure 5.2 presents five representative test predictions produced by the Stage-2 MobileNetV2 disease classifier. Each sub-figure shows the input leaf image together with the model's predicted disease category.

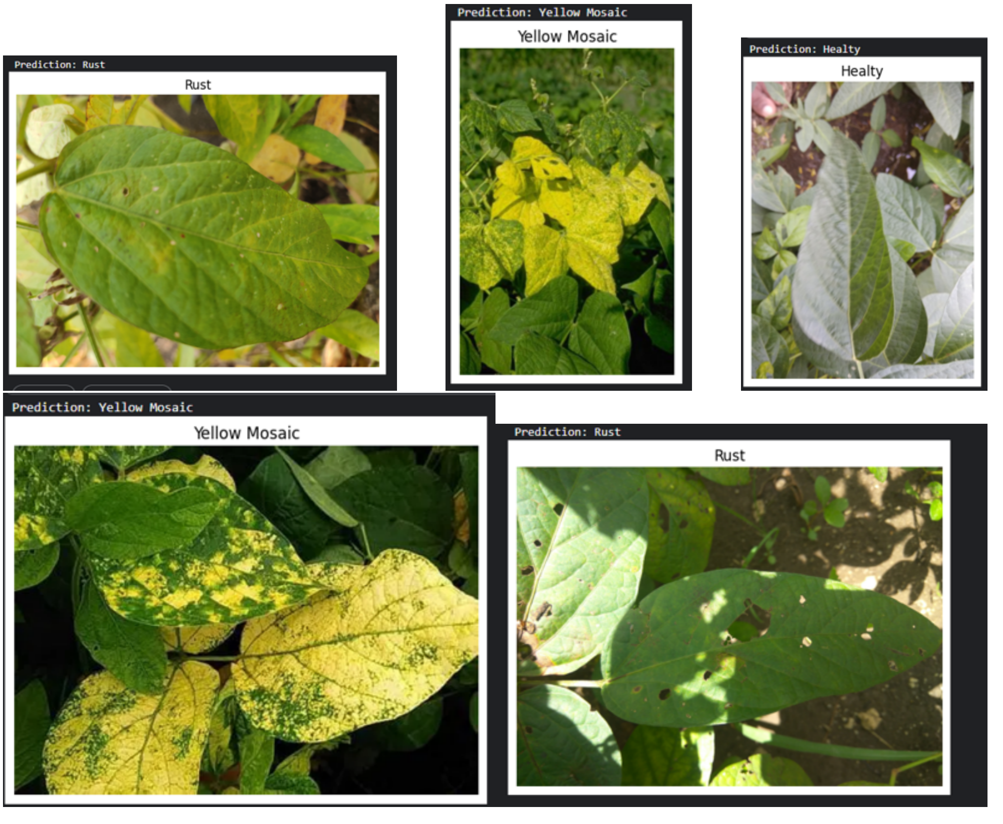

Figure 5.2: Stage-2 disease classifier: five representative test predictions. Each image already contains the model's prediction and ground truth label.

5.4 Challenges Encountered

1. **Dataset discovery and class harmonisation:** Identifying a single dataset covering all four target classes with consistent labelling required

significant evaluation time.

2. **Label inconsistencies and corrupted images:** A manual curation step was necessary before training could begin.

3. **Intermittent CUDA runtime issues:** Large-batch training encountered non-reproducible CUDA allocation errors, mitigated by reducing batch size and restarting with deterministic seeds.

CHAPTER 6

Conclusions and Scope for Future Work

6.1 Conclusions

This internship project proposed and validated a **two-stage deep learning pipeline** for soybean leaf disease detection consisting of:

1. A **ResNet-based leaf detector** achieving 99.07% validation accuracy to filter non-leaf inputs.

2. A **MobileNetV2-based disease classifier** achieving 86.32% test accuracy and 87.10% validation accuracy across four disease/health categories.

The pipeline addresses a critical practical limitation of single-stage classifiers: their fragility to out-of-distribution inputs in uncontrolled field environments. The sequential gating design eliminates spurious disease predictions, reduces unnecessary compute, and produces a system that is architecturally appropriate for edge and mobile deployment.

Both models employ transfer learning from ImageNet pre-trained weights, augmented with task-specific data augmentation, delivering strong baseline performance without requiring large-scale custom datasets. The two-month internship period provided sufficient time to design, implement, evaluate, and document both pipeline stages, establishing a solid foundation for production deployment and further refinement.

6.2 Scope for Future Work

1. **Dataset expansion:** Collecting field-captured soybean images under diverse lighting conditions, growth stages, and geographic settings will

improve robustness beyond curated laboratory data.

2. **Class-level refinement:** The Yellow Mosaic class (F1 = 0.69) needs targeted data collection, class-weighted loss, and mixup augmentation to boost recall.

3. **Confusion matrix analysis:** A rigorous per-class analysis will quantify systematic misclassification pairs (e.g., Yellow Mosaic $\leftrightarrow$ Rust) and guide future interventions.

4. **Lightweight localisation:** If downstream workflows require knowing the disease location on the leaf, integrating a YOLO-lite or EfficientDet model as an optional Stage 2b should be explored.

5. **Edge device prototype:** Deploying the pipeline (TFLite or ONNX) on a Raspberry Pi or Android device will provide real-world latency and power-consumption benchmarks.

6. **Broader disease coverage:** Extending the Stage-2 classifier to additional soybean diseases (e.g., Frogeye Leaf Spot, Bacterial Pustule) will increase the practical utility of the system.

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CHAPTER A

Training Metrics Summary

Table A.1: Complete Metrics Summary for Both Pipeline Stages

Stage	**Metric**	**Value**
Stage 1 (ResNet)	Validation Accuracy	99.07%
	YOLO Comparison	$\approx$94%
Stage 2 (MobileNetV2)	Test Accuracy	86.32%
	Macro Precision	0.8815
	Macro Recall	0.8632
	Macro F1 Score	0.8565
	Validation Loss	0.1173
	Validation Accuracy	87.10%

CHAPTER B

Hyperparameter Configuration

Table B.1: Full Hyperparameter Listing — Stage 2 MobileNetV2

Parameter	**Value**
Input size	$224 \times 224 \times 3$
Pre-trained backbone	MobileNetV2 (ImageNet)
Classifier head	Linear(1280, 4)
Optimiser	Adam
Learning rate	1×10^{-3} (default)
β_1, β_2	0.9, 0.999
Loss	CrossEntropyLoss
Batch size	32
Epochs	10
Data augmentation	RandomHorizontalFlip, RandomRotation
Normalisation mean	[0.485, 0.456, 0.406]
Normalisation std	[0.229, 0.224, 0.225]